

Smart Stethoscope System Design

Dual Analog MEMS Acoustic Sensing, PVDF Piezoelectric Vibration Path,
Low-Noise Analog Front-End, BLE Connectivity, and Battery Management

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1 Introduction

This document presents the proposed product-grade design approach for a smart stethoscope. The goal is to move beyond a basic microphone-based prototype and develop a professional biomedical sensing platform capable of clean heart and lung sound acquisition, direct diaphragm vibration sensing, contact-quality detection, battery monitoring, and mobile application connectivity.

The proposed system uses two high-performance Infineon analog MEMS microphones for acoustic sensing, one PVDF piezoelectric sensor for direct diaphragm vibration sensing, a low-noise analog front-end, a multi-channel audio ADC, a BLE microcontroller, and a complete Li-ion battery charging and fuel-gauge power management circuit.

2 Objective

The main objective is to design a reliable and scalable smart stethoscope hardware platform that can be used for high-quality physiological sound recording and future AI-based diagnosis.

The system is expected to provide:

- High-quality heart sound and lung sound acquisition.
- Better signal quality than basic electret microphone modules.
- Direct diaphragm vibration sensing using a PVDF piezoelectric film.
- Motion and contact-quality detection during recording.
- BLE-based communication with a mobile application.
- Battery charging, protection, and percentage monitoring.
- A professional hardware base for future AI-based classification.

3 Final System Approach

The proposed design uses a three-path sensing architecture.

3.1 Path A1: Main Acoustic Sensing

The first acoustic path uses the **Infineon IM68A130** analog MEMS microphone. This microphone will be placed inside the sealed acoustic chamber of the chest piece. It will be used as the main heart sound acquisition channel because it has strong low-frequency response, which is important for phonocardiogram signal capture.

3.2 Path A2: Reference Acoustic Sensing

The second acoustic path uses the **Infineon IM70A135** differential analog MEMS microphone. This will act as a secondary reference channel for high dynamic range acquisition, lung sound support, and noise comparison. It can also help improve signal reliability by comparing the main and reference acoustic channels.

3.3 Path B: Piezoelectric Vibration Sensing

The third path uses the **TE LDT0-028K PVDF piezoelectric film**. This sensor will be mechanically coupled to the diaphragm or backplate of the chest piece. It will directly sense vibration from the diaphragm, which can be useful for contact-quality detection, vibration analysis, and motion artifact rejection.

4 System Block Diagram

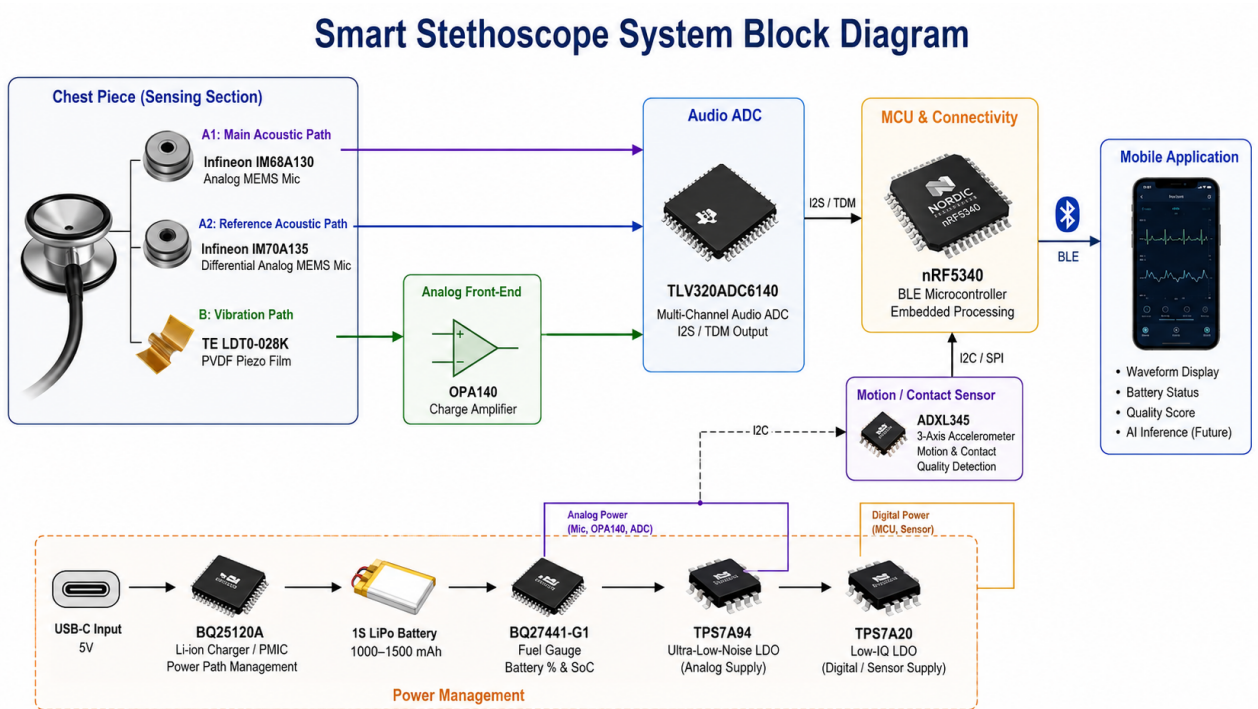


Figure 1: Proposed product-grade smart stethoscope system architecture.

5 Component Selection

Table 1: Main component list

Block		Component	Purpose
Main Sensor	Acoustic	Infineon IM68A130	Main low-frequency heart sound and PCG acquisition channel.
Reference Sensor	Acoustic	Infineon IM70A135	Secondary differential acoustic channel for reference, lung sounds, and noise comparison.
Piezoelectric Sensor	Sensor	TE LDT0-028K PVDF Film	Direct diaphragm vibration sensing and contact-quality support.
Piezo Front-End		TI OPA140	High-impedance charge amplifier for the PVDF piezoelectric sensor.
Audio ADC		TI TLV320ADC6140	Multi-channel ADC for digitizing microphone and piezo front-end signals.
Motion Sensor		ADXL345	Low-cost motion and contact-quality detection sensor.
Microcontroller		Nordic nRF5340	BLE communication, data handling, and embedded processing.
Charger/PMIC		TI BQ25120A	Li-ion battery charging and power management.
Fuel Gauge		TI BQ27441-G1	Battery percentage and state-of-charge monitoring.
Analog Regulator		TI TPS7A94	Ultra-low-noise supply for analog front-end and ADC.
Digital/Sensor Regulator		TI TPS7A20	Low-power regulated supply for MCU and sensor blocks.

6 System Working Process

The smart stethoscope will work as follows:

1. The diaphragm of the chest piece captures body sounds from the patient.
2. The IM68A130 microphone captures the main heart sound signal from the sealed acoustic chamber.
3. The IM70A135 microphone captures a secondary acoustic reference signal for noise comparison and lung sound support.
4. The PVDF piezoelectric film captures direct mechanical vibration from the diaphragm or backplate.
5. The OPA140 charge amplifier conditions the weak piezoelectric signal.
6. The TLV320ADC6140 digitizes all analog channels and sends the data to the nRF5340 through I2S/TDM.
7. The ADXL345 detects motion and contact-quality issues during recording.
8. The nRF5340 sends the processed data to the mobile app using BLE.
9. The mobile app displays waveform, battery percentage, quality score, and future AI-based diagnostic results.

7 Power Management Design

The device will use a 1S LiPo battery with a professional power management architecture. The analog and digital power rails will be separated to minimize noise coupling into the sensitive microphone and piezoelectric front-end.

- **USB-C Input:** Used for charging and optional debugging.
- **BQ25120A:** Handles Li-ion charging and power-path management.
- **BQ27441-G1:** Measures battery voltage, state of charge, and battery percentage.
- **TPS7A94:** Provides ultra-low-noise analog supply for the ADC, microphones, and charge amplifier.
- **TPS7A20:** Provides low-power supply for MCU and digital sensors.
- **1S LiPo Battery:** Recommended capacity is 1000–1500 mAh for compact size and good runtime.

8 PCB Design Strategy

The PCB should be designed as a 4-layer board for proper grounding, power integrity, and low-noise analog performance.

Recommended layer stack:

- **Layer 1:** Components and critical signal routing.
- **Layer 2:** Solid ground plane.
- **Layer 3:** Power planes and controlled routing.
- **Layer 4:** Secondary signal routing.

Important PCB design rules:

- Keep microphone and piezo traces short.
- Keep analog and digital sections separated.
- Use clean analog ground and connect carefully to system ground.
- Place TPS7A94 close to ADC and analog front-end.
- Keep BLE antenna area clear of copper and metal enclosure.
- Add test points for all power rails and ADC communication lines.
- Use shielding and proper mechanical grounding around the piezo front-end.

9 Estimated Cost

The following cost is an approximate estimate for the first prototype. Actual cost may vary depending on supplier, JLCPCB part availability, PCB size, assembly options, shipping, and customs.

9.1 Electronics BOM Estimate

Table 2: Estimated electronics BOM cost

Component	Part	Qty	Unit Cost	Total
Main analog MEMS microphone	IM68A130	1	\$1.64	\$1.64
Reference analog MEMS microphone	IM70A135	1	\$2.09	\$2.09
PVDF piezoelectric film	LDT0-028K	1	\$9.43	\$9.43
Piezo charge amplifier	OPA140	1	\$3.00	\$3.00
Multi-channel audio ADC	TLV320ADC6140	1	\$14.40	\$14.40
Motion/contact sensor	ADXL345	1	\$3.00	\$3.00
BLE microcontroller	nRF5340	1	\$9.46	\$9.46
Battery charger/PMIC	BQ25120A	1	\$4.06	\$4.06
Fuel gauge	BQ27441-G1	1	\$3.89	\$3.89
Ultra-low-noise analog LDO	TPS7A94	1	\$8.01	\$8.01
Low-power LDO	TPS7A20	1	\$0.35	\$0.35
USB-C, ESD, switch, connectors	Mixed	1 set	\$5.00	\$5.00
Crystals, antenna, RF matching	Mixed	1 set	\$8.00	\$8.00
Passives, filters, ferrites	Mixed	1 set	\$10.00	\$10.00
1S LiPo battery	1000–1500 mAh	1	\$4.00	\$4.00
Estimated Electronics Subtotal				\$92.33

9.2 JLCPCB Prototype Manufacturing Estimate

Table 3: Estimated JLCPCB PCB and assembly cost

Item	Estimated Cost
4-layer PCB fabrication, 5 pieces	\$10–25
SMT assembly setup fee, one side	\$25
Assembly and placement charges	\$20–40
Extended or special component handling	\$15–40
Shipping estimate	\$20–40
Estimated JLCPCB Manufacturing + Assembly	\$90–170

9.3 Total Prototype Budget

Table 4: Total estimated prototype budget

Category	Estimated Cost
Electronics BOM	\$92.33
JLCPCB PCB + assembly + shipping	\$90–170
Mechanical chest-piece and enclosure prototype	\$30–80
Extra parts, rework margin, and test connectors	\$30–50
Expected First Prototype Budget	\$250–400

Assuming an approximate exchange rate of:

$$1 \text{ USD} \approx 280 \text{ PKR}$$

The expected first prototype budget is:

$$\$250 \text{ to } \$400 \approx \text{PKR } 70,000 \text{ to PKR } 112,000$$

For safe planning, a budget of around **PKR 100,000 to PKR 125,000** should be kept for the first prototype cycle.

10 Development Plan

10.1 Phase 1: Design Finalization

- Finalize the custom chest-piece structure.
- Decide microphone chamber geometry.
- Decide PVDF film placement on diaphragm or backplate.
- Finalize schematic-level architecture.

10.2 Phase 2: Schematic Design

- Design IM68A130 and IM70A135 analog microphone circuits.
- Design OPA140-based charge amplifier for PVDF sensing.
- Connect all analog paths to TLV320ADC6140.
- Design nRF5340 BLE microcontroller section.
- Add BQ25120A charger, BQ27441-G1 fuel gauge, and LDO power rails.

10.3 Phase 3: PCB Layout

- Design a 4-layer PCB.
- Separate analog and digital domains.
- Keep low-noise analog traces short and shielded.
- Maintain proper BLE antenna clearance.
- Add test points for debugging and validation.

10.4 Phase 4: JLCPCB Manufacturing

- Generate Gerber files, BOM, and CPL files.
- Check JLCPCB component availability.
- Use JLCPCB SMT assembly where possible.
- Hand-solder or consign components that are not available in JLCPCB library.

10.5 Phase 5: Hardware Testing

- Verify all power rails.
- Test battery charging and battery percentage reading.
- Test ADC communication with the nRF5340.
- Test both acoustic microphone channels.
- Test PVDF vibration channel.
- Test ADXL345 motion/contact detection.

10.6 Phase 6: Signal Validation

- Record heart sounds from different chest positions.
- Compare IM68A130, IM70A135, and PVDF signal quality.
- Evaluate noise, signal strength, repeatability, and contact artifacts.
- Finalize filter settings and sampling rate.

10.7 Phase 7: Mobile App Integration

- Send waveform data to mobile app through BLE.
- Display real-time waveform.
- Display battery percentage.
- Add quality score based on acoustic, piezo, and accelerometer data.
- Prepare data pipeline for future AI-based classification.

11 Expected Outcome

At the end of the first prototype cycle, the expected outcome will be a professional smart stethoscope hardware platform with:

- Dual analog MEMS microphone acoustic recording.
- PVDF piezoelectric diaphragm vibration sensing.
- Motion and contact-quality detection.
- Multi-channel low-noise audio acquisition.
- BLE-based mobile app connectivity.
- Battery charging and battery percentage monitoring.
- A scalable hardware platform for AI-based diagnosis.

12 Conclusion

The proposed design converts the smart stethoscope from a basic prototype into a professional biomedical sensing system. By using dual Infineon analog MEMS microphones, a PVDF piezoelectric vibration path, a low-noise analog front-end, a dedicated multi-channel ADC, BLE communication, and proper battery management, the system becomes suitable for serious product development and future AI-based medical signal analysis.